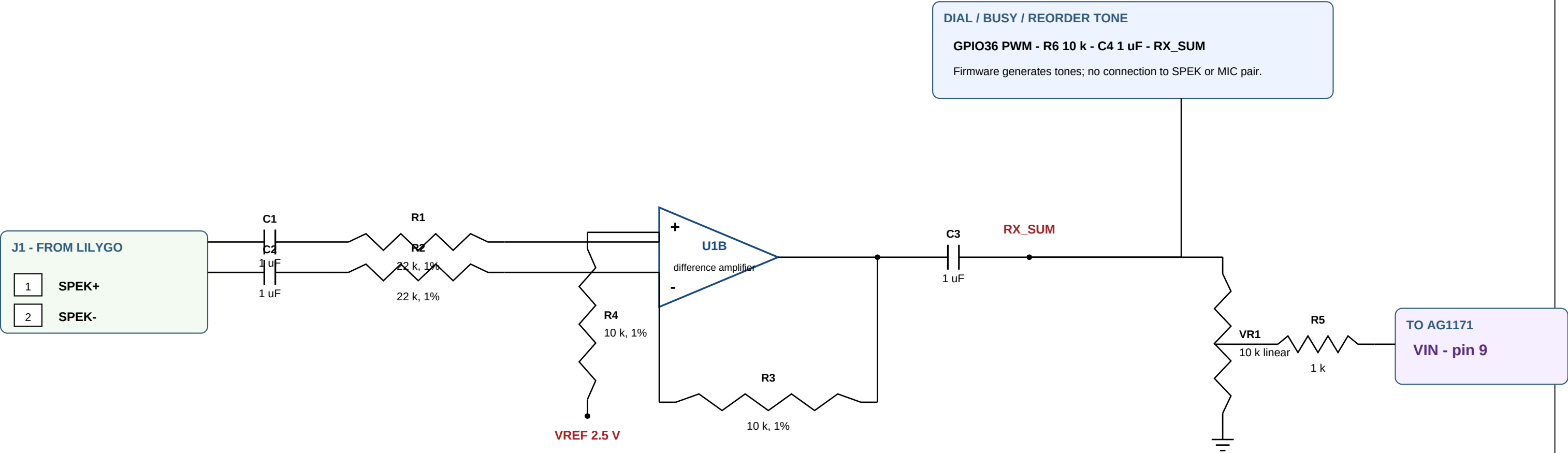


CELL -> ROTARY: LilyGO SPEK differential output to AG1171 VIN pin 9



EXACT U1B CONNECTIONS

SPEK- -> C2 -> R2 -> U1B inverting (-) input.
U1B output -> R3 -> the same inverting (-) input.
SPEK+ -> C1 -> R1 -> U1B non-inverting (+) input.
VREF -> R4 -> the same non-inverting (+) input.
R3 and R4 are therefore not in parallel.

TRANSFER AND STARTUP

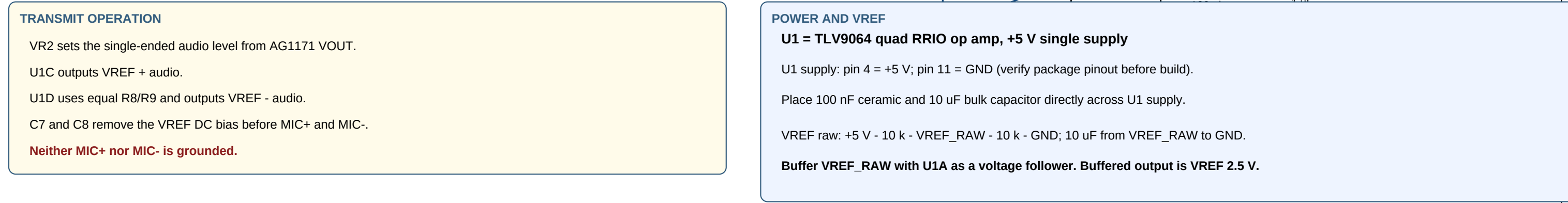
Because $R3/R2 = R4/R1 = 10k/22k$:

U1B OUT = VREF + 0.455 x (SPEK+ - SPEK-)

Set VR1 at minimum before first test: wiper-to-ground approximately 0 ohms.
Increase VR1 slowly during a connected call.

PAGE 1 NOTES

- 1. Neither SPEK+ nor SPEK- is connected to ground. Both inputs are AC-coupled.
- 2. U1B is one section of a TLV9064 quad rail-to-rail op amp on +5 V.
- 3. C1-C4 may be 1 uF film or low-leakage X7R. R1-R4 should be 1% matched values.
- 4. This page replaces the receive transformer only. AG1171 ring/hook circuitry is unchanged.



1. Start VR2 near minimum and raise it only until the far-end transmit level is adequate.
2. **The values shown are a conservative first-pass interface, not yet bench-validated on this exact LilyGO board.**
3. Verify LilyGO connector polarity and TLV9064 package pinout before soldering.
4. Keep receive and transmit traces physically separated; use a quiet star ground near AG1171/LilyGO common ground.